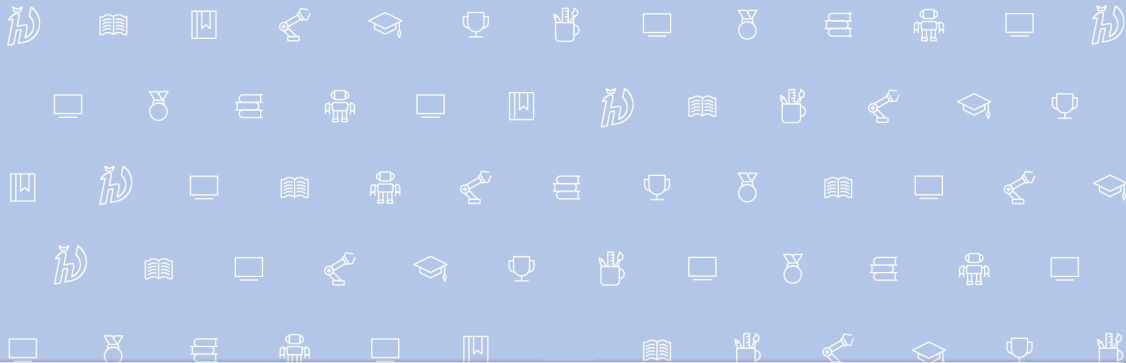




Artificial Intelligence Software, Lecture 3

AI Paper Pipeline: Writing, Reading, and Reviewing with AI



Center for
Cognitive
Modeling



Outline

1. Introduction
2. Lifecycle of a Paper
3. Searching for Literature
4. Reference Management
5. How to Read Papers
6. Generating Core Part
7. Proofreading
8. Auto-review
9. Detection of Generated Text

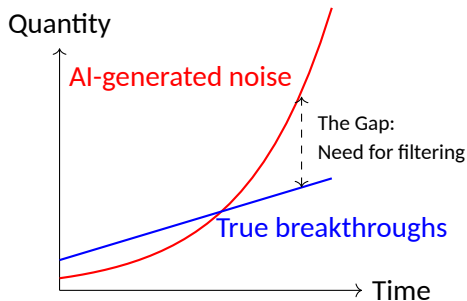


Introduction: The Researcher in the AI Era

- **Role Shift:** From generating text to generating meaning. AI is a tool, not the author.
- **Knowledge vs. Data:** AI models reflect existing ideas. New knowledge comes from researchers.
- **The Noise Problem:**
 - Easier to generate text → Flood of low-quality papers.
 - Speed vs. Quality trade-off.
- **Value of True Research:** High-quality, reproducible work stands out more than ever.



Publish or Perish: New Reality



- **Reproducibility** is the new competitive advantage.
- **Open Code** is a sign of respect.
- **Transparency** builds trust.



Responsibility and Reputation

- **Author's Responsibility:**
 - Setting the research problem.
 - Conducting experiments.
 - Interpreting results.
 - Ethics and safety.
- **Reputation:** AI can write, but it cannot own the results.
- **Expertise:** You are the navigator in the information noise.

Paper Lifecycle Overview





The Process Levels

1. Researcher's Work:

- Idea generation & Experiments.
- Drafting (iterative process).

2. Collaborative Work:

- Internal review (Colleagues).
- AI pre-review (sanity check).

3. Community Work:

- Submission (Conferences, Workshops, Journals).
- External Review & Rebuttal.

Preparation for Submission

- **Deadlines:** Planning is key (Abstract deadline vs Full paper deadline).
- **Venue Selection:** ICLR, NeurIPS, ICML, CVPR, AAAI, etc.
- **Formal Requirements:**
 - Page limits (Strict!).
 - Templates (LaTeX).
 - Anonymity (Double-blind review).

Submission Systems

OpenReview.net (ICLR, NeurIPS)

OpenReview.net Search OpenReview... Notifications Activity Tasks Oleg Bulichev

Open Peer Review, Open Publishing, Open Access, Open Discussion, Open Recommendations, Open Directory, Open API, Open Source, Donate

News

- Survey Finds Broad Support for Greater Openness in AI Peer Review Feb 13, 2026
- A Message from AI Research Leaders: Join Us in Supporting OpenReview Dec 19, 2025
- Initial Analysis of OpenReview API Security Incident Dec 01, 2025

[View all OpenReview news](#)

Your Active Consoles

- AAAI 2026 Workshop WoMAPF Authors

Active Venues

- TMLR
- Computo

Open for Submissions

- SolidProject SoSy 2026 Privacy Session
Due 19 Feb 2026, 01:59 Moscow Standard Time
- ACM CHI 2026 Workshop BiAlign
Due 19 Feb 2026, 02:59 Moscow Standard Time
- TPDP 2026
Due 19 Feb 2026, 02:59 Moscow Standard Time

Microsoft CMT (CVPR, ICCV)

Submissions Select Your Role: Author

Author Console

1 - 1 of 1 Show Hide Clear all Filters

Paper ID	Title	Files	Status	Actions
17	D Paper Show abstract	Submission files: Scientific Paper .pdf	Revision View Reviews View Meta-Reviews	Revision: Upload Revision



Workshops and Journals

Workshops:

- Work in progress.
- Networking & Feedback.
- Specific narrow topics.
- Less strict than main conference.

Journals:

- Archival value.
- Longer review cycle (months/years).
- More detailed description.



Reference Materials

- Conference websites (NeurIPS, ICLR, etc.) for requirements.
- [OpenReview.net](https://openreview.net)
- Microsoft CMT



Searching for Literature

- **Initial Search:**
 - **Semantic Scholar:** standard for AI papers.
 - **Asta (Ai2):** emerging tool.
 - **Deep Research Agents:** perplexity, etc.
 - **Scholar inbox:** AI-powered literature search.
- Don't just rely on Google. Use academic-specific engines.



Apps for Literature Search



A scholarly research assistant that combines literature understanding and data-driven discovery. Asta uses 108M+ abstracts and 12M+ full-text papers to find, summarize, and analyze scientific evidence. A project from [AI2](#).

Find papers

Generate a report

Analyze data

Ask a scientific review question



See what's possible. Choose an example to try in the prompt box:

Compare two theories or frameworks · Extract best practices from multiple papers · Understand the tradeoffs behind a specific method · Survey different approaches for a research task

Asta (AI2)



SLAM

Search

Open Fullscreen

About 34,700 results for "SLAM"

Fields of Study

Date Range

Has PDF

Author

Journals & Conferences

Sort by Relevance

1 2 3



Abdolkarim Azarbayjani
2,500 Publications · 2,500 Citations · Chemistry



Stan
10 Publications · 10 Citations · Agriculture and Food Sciences



K. Sani
10 Publications · 10 Citations · Medicine

Show All Authors

ORB-SLAM2: An Open-Source SLAM System for Monocular, Stereo, and RGB-D Cameras
Raul Mur-Artal · J. D. Trulls · Computer Science, Engineering · IEEE Transactions on Robotics · 30 October 2015

SLAM: ORB-SLAM2, a complete simultaneous localization and mapping (SLAM) system for monocular, stereo and RGB-D cameras, including map reuse, loop closing and relocalization capabilities, is presented, being in most cases the most accurate SLAM solution. Expand

14,519 · 10 PDF · 5,000 · 10 Cit · 10 Cit

ORB-SLAM2: An Accurate Open-Source Library for Visual, Visual-Inertial, and Multimap SLAM

C. Campos · Richard Dierker · J. Rodriguez · Juan M. M. Montiel · Juan D. Trulls

Computer Science, Engineering · IEEE Transactions on Robotics · 29 July 2016

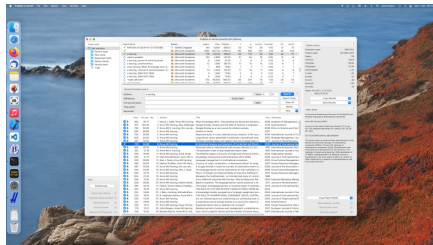
SLAM: This article presents ORB-SLAM2, the first system able to perform visual, visual-inertial and multimap SLAM with monocular, stereo and RGB-D cameras, using pin-hole and fisheye lens models, resulting in real-time robust operation in varied and large, indoor and outdoor environments. Expand

14,519 · 10 PDF · 5,000 · 10 Cit · 10 Cit

Semantic Scholar

Publish or Perish (Application)

- **What is it?** A software program that retrieves and analyzes academic citations.
- **Key Features:**
 - Uses Google Scholar, Scopus, Web of Science.
 - Calculates h-index, g-index metrics.
 - Detailed citation analysis for authors or journals.
 - Helps identify "Must Read" classic papers by citation count.



- Build collections of papers.
- AI learns what you like and improves recommendations.
- Interactive visualization of author/paper networks.
- Infinite feed of related work.





Structuring & Information Extraction

- **Elicit:**
 - "Research Assistant".
 - Extract data from papers into a structured table.
 - "What is the NMAE score for X?" across 10 papers.
- **NotebookLM:** RAG system for synthesis.

Elicit Recent Library Alerts

Q Instruction following in minecraft

Sort: Most relevant Search Filters + Column Export UPGRADE Save to library Get 2 more full-texts Open

Paper	Summary	training data	rl use
☐ MineDreamer: Learning to Follow Instructions via Chain-of-Imagination for Simulated-World Control Enshen Zhou, Yiran Qin, Zhen-fei Yin, Yuzhou Huang, Ruimao Zhang, and 3 more arxiv.org, 2024, 40 citations Full text	MineDreamer, an embodied agent in Minecraft, uses a Chain-of-Imagination mechanism to follow instructions steadily, outperforming existing baselines by nearly doubling performance in single and multi-step instructions.	The training data is the Goal Drift Dataset, which includes 500,000 triplets of current observation, future goal imagination, and instruction, derived from the OpenAI Contractor Gameplay Dataset and the MineRL BASALT 2022 dataset.	The "rl use" in the paper refers to the reinforcement learning fine-tuning of the VPT model, specifically the VPT(l) variant, which is used as a baseline in the experiments.
☐ STEVE-1: A Generative Model for Text-to-Behavior in Minecraft Shalev Lifshitz, Kaven Paster, Nats, Chan, Jimmy Ba, Stella A. McIlraith Neural Information Processing Systems, 2023, 85 citations Full text	STEVE-1, a generative model, follows short-horizon open-ended text and visual instructions in Minecraft, completing 12 of 13 tasks, using a methodology inspired by unCLIP and leveraging pretrained models like VPT and MineCLIP.	- Contractor gameplay dataset: 35M frames - VPT-generated gameplay dataset: 15M frames - Manually gathered text instructions: 2,000 text-video pairs - Augmented dataset: Additional text-gameplay pairs using MineCLIP and OpenAI API	The paper discusses the use of reinforcement learning (RL) techniques such as behavioral cloning, hindsight relabeling, and supervised RL in the context of sequential decision-making in Minecraft. These methods are used to fine-tune the VPT model to follow instructions without relying on large datasets of instruction-labeled trajectories.
☐ GROOT: Learning to Follow Instructions by Watching Gameplay Videos Shaofei Esi, Bewei Zhang, Zihao Wang, Xiaojian Ma, Anji Liu, and 1 more International Conference on Learning Representations, 2023, 34 citations Full text	GROOT, an agent using reference videos as instructions, closes the human-machine gap in Minecraft with a 70% winning rate over the best baseline.	The training data includes a mixture of Minecraft gameplay videos and offline trajectories from the contractor data, excluding metadata.	- Dye and sheep wool: 100% - Enchant sword: 100% - Smelt food: 100% - Use bow: 100% - Sleep: 100% - Lead animals: 100% - Build snow golems: 50% - Build obsidian: 50%

Automated Extraction Columns: Methodology, Dataset, Metrics, Limitations.



The MCP Pipeline (Advanced)

Automation with MCP (Model Context Protocol):

1. **Prompt:** Define strict search criteria.
2. **AI Agent + Google Scholar MCP:**
 - Agent queries Scholar directly via MCP.
 - Extracts Abstract & DOI.
 - Saves to Markdown.
3. **RAG Integration:** Feed md/pdfs to NotebookLM.
4. **Generation:** Synthesize Technical Report.



Reference Materials

- Google Scholar MCP Server
- ResearchRabbit
- Publish or Perish App
- Semantic Scholar
- Elicit.com
- Litmaps
- Perplexity
- Asta (Ai2)
- Scholar Inbox



Zotero: The Knowledge Hub

- **Why Zotero 8?** Modern UI, native speed, built-in PDF reader.
- **Browser Connector:** One-click save (PDF + Metadata) from ArXiv, IEEE, etc.
- **Built-in Features:**
 - Automatic file renaming based on parent metadata.
 - Extracting annotations to notes directly.
- **Note Templates:** Configurable templates regarding your workflow.



Essential Plugins

- **Better BibTeX:** Generates stable Citation Keys (e.g., bulichev2024ais) for LaTeX/Overleaf.
- **Zotero SciPDF:** Automatically downloads PDFs via Sci-Hub if paywalled.
- **Better Notes:** Advanced knowledge management, markdown support, and bi-directional linking within Zotero.



Teamwork & BibTeX Generation

- **Group Libraries:**

- Share papers, PDFs, and notes with colleagues.
- Everyone sees the same PDF annotations.

- **BibTeX Workflow:**

- Right-click Collection → Export Collection → Better BibTeX.
- Select "Keep updated" to auto-sync .bib file.
- **Overleaf:** Link the .bib file via URL or upload manually.
- **Citation Keys:** Ensure consistency (e.g., [auth][year]).



Obsidian Integration Workflow

- **Goal:** Turn PDF highlights into permanent notes in your "Second Brain".
- **Tool:** *Citations* plugin (Obsidian).
- **Process:**
 1. Annotate PDF in Zotero.
 2. Use "Add Item Note from Annotations".
 3. Open Obsidian → "Insert Literature Note".
 4. Template auto-fills Title, Authors, Abstract, and your Zotero highlights.



Zotero GUI

The image displays the Zotero GUI interface. On the left is a sidebar with a file explorer showing folders like 'My Library', 'Books', 'Copy of ST MS SLA...', 'Interesting links', 'kek', 'protective_shell_fo...', 'Tablet Files (modifi...', 'My Publications', 'Duplicate Items', 'Unfiled Items', and 'Trash'. Below these are 'Group Libraries' and 'Protective_shell_fo...' with sub-folders 'LR', 'LR papers', 'Duplicate Items', 'Unfiled Items', and 'Trash'. At the bottom of the sidebar are 'SLAMM' and 'ST MS SLAM' with sub-folders 'Duplicate Items', 'Unfiled Items', 'Trash', 'MIT Spark', and 'Duplicate Items'. A legend at the bottom left indicates: 'Not overviewed' (green dot), 'Survey' (orange dot), and 'Dynamic Mapping' (blue dot).

The main pane shows a list of publications with columns 'Title' and 'Creator'. The selected item is 'A survey of state-of-the-art on visual SLAM' by Abaspur Kazerouni et al.

Title	Creator
> 3D LIDAR SLAM : A survey	Zhang et al.
> A comprehensive overview of core modules in visual SLAM framework	Cai et al.
> A Comprehensive Survey of Visual SLAM Algorithms	Macario Barros et al.
> A portable three-dimensional LIDAR-based system for long-term and wide-area people behavior measurement	Koide et al.
> A Review of Multi-Sensor Fusion SLAM Systems Based on 3D LIDAR	Xu et al.
> A survey of state-of-the-art on visual SLAM	Abaspur Kazerouni et al.
> A Survey of Visual SLAM in Dynamic Environment: The Evolution From Geometric to Semantic Approaches	Wang et al.
> A Survey on Reinforcement Learning Applications in SLAM	Tezerjani et al.
> An Overview on Visual SLAM: From Tradition to Semantic	Chen et al.
> BEVPlace++: Fast, Robust, and Lightweight LIDAR Global Localization for Unmanned Ground Vehicles	Luo et al.
> GLIM: 3D Range-Inertial Localization and Mapping with GPU-Accelerated Scan Matching Factors	Koide et al.
> Hydra-Multi: Collaborative Online Construction of 3D Scene Graphs with Multi-Robot Teams	Chang et al.
> Hydra: A Real-time Spatial Perception System for 3D Scene Graph Construction and Optimization	Hughes et al.
> Khronos: A Unified Approach for Spatio-Temporal Metric-Semantic SLAM in Dynamic Environments	Schmid et al.
> Kimera: an Open-Source Library for Real-Time Metric-Semantic Localization and Mapping	Rosinol et al.
> ORB-SLAM3: An Accurate Open-Source Library for Visual, Visual-Inertial and Multi-Map SLAM	Campos et al.
> PHD thesis lit review	
> PRISM-TopoMap: Online Topological Mapping with Place Recognition and Scan Matching	Muravyev et al.
> SLAM for Visually Impaired People: A Survey	Baird et al.
> Visual-localization-appearance-semantics-and-depth-matters-CAS-Data-Science-and-Machine-Learning	

The right pane shows the details for 'A survey of state-of-the-art on visual SLAM'. It includes a 'Title Translation' section and a 'Bibliography' section. The 'Bibliography' section lists the following information:

Item Type	Journal Article
Title	A survey of state-of-the-art on visual SLAM
Author	Abaspur Kazerouni, Iman
Author	Fitzgerald, Luke
Author	Dooley, Gerard
Author	Toal, Daniel
Publication	Expert Systems with Applications
Publisher	
Place	
Date	11/2022
Volume	205
Issue	
Section	
Part Number	
Part Title	
Pages	117734
Series	
Series Title	
Series Text	
Journal Abbr	Expert Systems with Applications
DOI	10.1016/j.eswa.2022.117734
Citation Key	abaspurkazerouni_Survey_2022
URL	https://linkinghub.elsevier.com/retrieve/pii/S0957417422010156
Accessed	1/13/2025, 5:00:33 PM
PMID	
PMCID	
ISSN	09574174



Reference Materials (Zotero)

- <https://www.zotero.org/>
- Zotero SciPDF (Zotero 7)
- Better Notes Plugin
- Better BibTeX Plugin



Reading Strategy with AI

Goal: Process information efficiently without losing depth.

- Using AI assistants (LLMs) to interact with papers.
- Not just summarizing, but querying.



Phase 1: Context & Hallucinations

- **Context Loading (RAG):** Use tools like NotebookLM or custom RAG pipelines to "chat" with the PDF.
- **Hallucination Check:**
 - Verify specific claims.
 - Ask for page numbers/citations in the response.
 - *"Where exactly does the paper state X?"*



Phase 2: The Truth Seeker

Generate Key Research Questions:

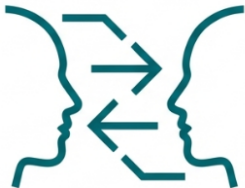
- Instead of "Summarize this", ask:
 - "What is the specific gap this paper fills?"
 - "What are the limitations explicitly mentioned by authors?"
 - "Compare method X with baseline Y based on the results tables."



Phase 3: Deep Analysis

- **Analyze Experiments:**
 - Check experimental setup rigor.
 - Metrics used (and why).
- **Extract Insights:**
 - What worked? What didn't?
 - Why did they choose this architecture?

Secret Technique: Author Debates



Copy This Prompt

Представь, что у нас есть две статьи: Article A и Article B. Ты уже получил их структуру, ключевые идеи и цитаты на основе предыдущего анализа. Сейчас мне нужно, чтобы ты смоделировал научные дебаты между их авторами, строго опираясь на текст статей.

Сначала выступают «авторы» Article A и критически рассматривают Article B. Пусть они укажут, какие аспекты мотивации, постановки задачи, выбора метрик, экспериментального сетапа или интерпретации результатов в Article B вызывают у них сомнения. Просьба формулировать критику в форме реплик, которые могли бы быть произнесены на реальном научном семинаре, и по возможности сопровождать ключевые замечания ссылками на секции, таблицы или рисунки Article B.

Затем дай ответ «авторов» Article B на эту критику. Пусть они объяснят, почему выбрали именно такую формулировку задачи, методику, экспериментальный дизайн или интерпретацию результатов, и укажут, в чём они видят слабые стороны Article A. Также по возможности опирайся на конкретные формулировки и данные из текста Article A.

Важно: не добавляй новых фактов, которых нет в статьях. Если какой-то аргумент нельзя обосновать цитатой или явным следствием из текста, обозначь его как гипотетический и минимизируй количество таких допущений.

Why: Forces the model to engage with the logic of the research, not just the text.



NotebookLM as a tool of avoiding hallucinations

The screenshot displays the NotebookLM interface with a notebook titled "SLAM and Fiducial Marker Integration Approaches". The interface is divided into three main sections: "Источники" (Sources), "Чат" (Chat), and "Студия" (Studio).

Источники (Sources): This section on the left allows users to add and manage sources. It includes a search bar "Найдите новые источники в интернете" and a list of selected sources:

- Интернет
- slam_fiducial_research.md
- Интеграция SLAM и fiducial-маркеров_обзор под...
- Найди научные источники и репозитории на github...

Чат (Chat): The central chat area contains a message from the AI assistant:

Источники выделяют такие инструменты, как **TagSLAM** и **RTAB-Map**, которые позволяют эффективно объединять данные с лидаров, колесных энкодеров и RGB-D камер. В текстах также рассматриваются специализированные решения, например **LIDARtag**, позволяющие распознавать маркеры непосредственно через лидарное облако точек. Предложенные стратегии включают создание иераргических карт, где **fiducial-метки** выступают в роли жестких ориентиров для замыкания циклов. В целом, данные публикации служат практическим руководством по построению надежных систем локализации для автономных роботов.

Below the message are interaction buttons: "Сохранить в заметке", "Лайк", and "Комментарий".

Further down, there are three prompts in a light blue box:

- В чем разница между фильтрацией и Graph SLAM для сброса ошибок?
- Как интегрировать Apriltags в процесс построения 3D карты лидаром?
- Какие готовые пакеты ROS лучше всего объединяют LIDAR и маркеры?

A date separator indicates "воскресенье, 7 декабря".

Below the prompts is a light blue box with the text: "каким образом на эталонной лидарной карте установить fiducial".

At the bottom, a note states: "Установка (или локализация) fiducial-маркеров на эталонной лидарной карте может осуществляться несколькими способами, в зависимости от того, строится ли карта одновременно с увеличением масштаба или маркеры привязываются к уже готовой карте."

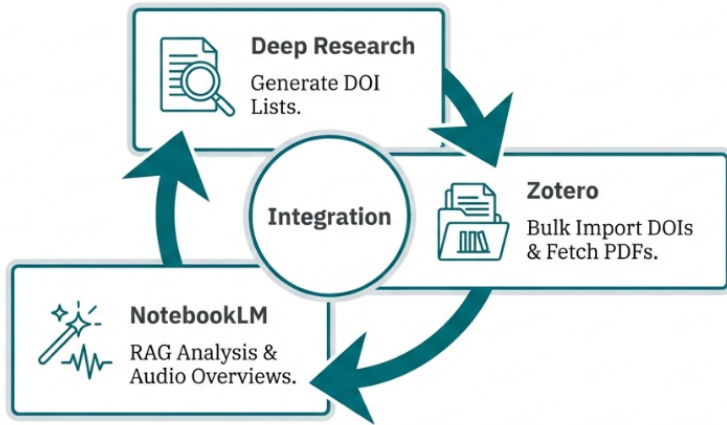
Студия (Studio): The right-hand section shows a grid of cards for different content types:

- Аудиопересказ
- Видеопересказ
- Ментальная карта
- Отчеты
- Карточки
- Тест
- Infographics (Infografika)
- Презентация
- Таблица данных

Below the grid, a list of documents is shown:

- Fiducial Markers For Robust SLAM (3 источника · 72 дня назад)
- SLAM и Маркеры Карта Решений (2 источника · 73 дня назад)
- Fiducial SLAM Precision Navigation Strategies (3 источника · 73 дня назад)
- Технический меморандум: Сравнительный... (Анализ решений · 3 источника · 73 дня назад)

Your workflow





Reference Materials

- NotebookLM (Google).



From Plan to Text

Method 1: Plan → Connected Text

- Step 1: Create a detailed hierarchical plan (Chapters, Sections, Bullet points).
- Step 2: Use LLM to expand bullet points into paragraphs.
- **Crucial:** Freeze the structure. Don't let LLM wander.



3 Levels of Text Work

To write clear and structured text, operate on three distinct levels:

1. **Architecture Level:** The overall flow, chapter structure, and logic of the entire document.
2. **Paragraph Level:** The logical unit of thought. Structure within a block.
3. **Sentence Level:** Grammar, style, flow, vocabulary (Polishing).



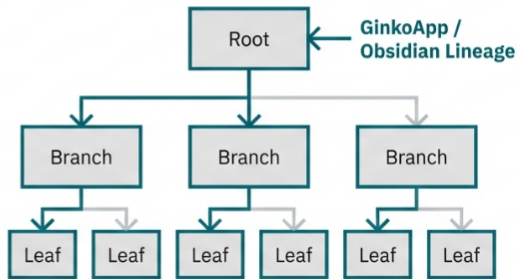
Rules and tools for focusing

The Golden Rule

One Idea =
One Paragraph.

If a paragraph has two ideas,
split it. If an idea spans two
paragraphs, merge it.

Block-Based Writing





Tools for Block Writing

1. GinkoApp:

- Tree-based writing tool.
- Visualizes the hierarchy of thoughts.

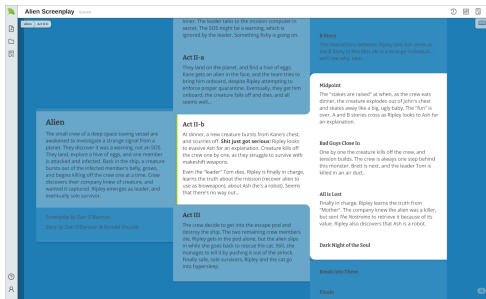
2. Lineage (Obsidian Plugin):

- "The Gold Standard" for local workflow.
- Markdown based.
- Works inside Obsidian.
- Easy export to LaTeX/Pandoc.

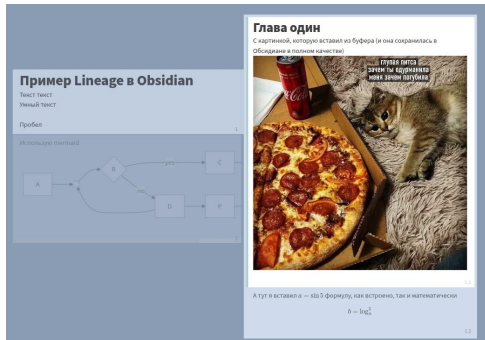


Block Writing in Action

GinkoApp Interface



Lineage (Obsidian)





From Draft to Text

Method 2: Draft → Connected Text

- Step 1: Write a "Brain Dump" (rough draft, messy English/Russian).
- Step 2: Ask LLM to "Rewrite in Academic English, maintaining the flow".
- Better for preserving YOUR original ideas.



Methodological Tricks

- **Reduce Cognitive Load:** Don't ask for "Outcome + Formatting + Style" in one go. Split tasks.
- **Iterative Refinement:**
 - Draft → Critique → Edit.
- **Context Management:** Provide definitions of symbols/terms to keep consistency.



Meta-Prompting (For the Lazy)

- Instead of writing the prompt yourself, ask a stronger model to write it.
- **Workflow:**
 1. Tell DeepSeek/Opus: "I want to generate a section about X. Write a detailed System Prompt for GPT-4 to do this perfectly."
 2. Use the generated prompt.

Image Generation for Papers

- **Flux:** Good for diagrams and schematic illustrations.
- **DALL-E 3:** Integrated, follows instructions well, but can be "cartoony".
- **Use Case:** Visual Abstracts, concept diagrams.
- **Pro Tip:** Generate parts of diagram and assemble in Figma/TikZ.



Reference Materials

- GinkoApp
- Obsidian Lineage Plugin
- Using Ginko to Do the Single Method of Academic Writing
- Flux (Black Forest Labs)
- DALL-E 3 (via ChatGPT)



Proofreading: Human vs AI

- **AI Proofreading:** Good for grammar, flow, typo detection.
- **Human Proofreading:** Essential for logic, nuance, and "vibe".
- **Strategy:** AI Check → Human Edit → AI Final Polish.
- Use "Track Changes" mode in prompts ("Show me what you changed").



Proofreading Interface Example (Grammarly)

Untitled document

Goals

91 Overall score

The experimental results **shown** a good performance of the distance computation algorithm in images taken with a low-cost camera, under uncontrolled conditions of illumination. The maximum error obtained was less than 0.9%, compared to a calibrated stereo vision system. Moreover, the system runs faster than the stereo system.

Review suggestions

Write with generative AI

Check for AI text & plagiarism

Review suggestions 3

Correctness

Clarity

Engagement

Delivery

★ Pro suggestions 1

Correctness · Change the form of the verb ⓘ

...experimental results **shown** **showed** a good performance...

Accept

Dismiss

...

Improve your text

The experimental results shown a good...



Title Engineering

- Title is the marketing of your paper.
- Use AI to generate 20 variations.
- Mix and Match: Combine the best parts.
- Keywords: Ensure SEO-friendly terms are in the title.



Pre-submission Checklist

1. Figures are readable in Black & White?
2. Captions explain the figure fully?
3. Citations are correct?
4. Formatting compliance?
5. Spellcheck done?



AI as Reviewer

Goal: Get feedback BEFORE the actual reviewers destroy you.

- **Tools:**
 - **cspaper:** Specializes in AI conferences (NeurIPS, ICLR).
 - **paperreview.ai**
- Simulate the peer-review process.



Structure of an AI Review

- **Summary:** Does the AI understand your paper? (If not, rewrite).
- **Strengths & Weaknesses:** Crucial feedback.
- **Questions:** What is unclear?
- **Rating:** (e.g., 1-10 scale).
- **Confidence:** (Low confidence = confusing paper).



Warning: Limitations

- **Missing Related Work:** AI often hallucinates papers here. Verification is mandatory.
- **Bias:** Models may prefer certain writing styles.
- **Use it for constructive critique, not for the score.**



Automated Review Interface



← Back

YOUR REVIEW

Desk Rejection

Assessment:

Paper Length

Topic Compatibility

Minimum Quality

Prompt Injection and Hidden

Manipulation Detection

Expected Review Outcome:

Review

Paper Length

Pass

Topic Compatibility

Pass The paper focuses on Multi-Agent Pathfinding (MAPF) using Imitation Learning and Transformers, which fits well within the "Multiagent Systems" and "Machine Learning" tracks of AAAI.

Minimum Quality

Pass The paper is complete, technically sound, and presents a clear methodology with extensive experiments.

Prompt Injection and Hidden Manipulation Detection

Pass No hidden prompts or manipulation detected.

Expected Review Outcome:

Review

The authors propose MAPF-GPT, a decentralized multi-agent pathfinding solver trained via imitation learning on a massive dataset of 1 billion expert observation-action pairs generated by LaCAM. The approach utilizes a transformer-based architecture to predict agent actions from local observations, employing a novel tokenization scheme that encodes map features and agent states. The paper demonstrates that this data-driven approach outperforms state-of-the-art learnable solvers like SCRIMP and DCC in terms of success rate and runtime, while exhibiting zero-shot generalization to unseen map types.

Pros and cons

Pros

- **Significant Dataset Contribution:** The creation and utilization of a 1 billion observation-action pair dataset (Section 3.2) is a commendable effort. Generating such a large-scale resource using a strong expert like LaCAM provides a solid foundation for this work and potentially for the broader community if released.
- **Strong Empirical Performance:** The results in Figure 3 are persuasive. MAPF-GPT-85M consistently outperforms SCRIMP and DCC across 'Random' and 'Mazes' maps. Specifically, on 'Mazes' with 32 agents, MAPF-GPT maintains a success rate above 80% while SCRIMP drops significantly below 60%. The tight confidence intervals (shaded areas) indicate robust performance.



Reference Materials

- <https://cspaper.org/>
- paperreview.ai



The Detection Problem

- **ICLR Case:** Massive lazy reviewing with AI.
- 21% of reviews were suspected to be AI-generated.
- **Why it matters:** Low effort reviews hurt science.



Signs of Generated Text (Manual)

- **Vocabulary:** Overuse of "delve", "underscore", "tapestry", "seamless", "landscape".
- **Structure:** Perfect, monotonous paragraph lengths.
- **Lack of Specifics:** Vague claims ("method is robust") without data ("method achieved 95%").



Technical Detection

- **Unicode Artifacts:** AI might use weird symbols.
- **Services:**
 - GPTZero, Pangram, Originality.ai.
- **Problem:** High False Positive rate.
- Good papers can be flagged as AI. Don't trust detectors 100%.



Reference Materials

- Pangram Labs
- GPTZero



AI is the accelerator. You are the driver. Keep your hands on the wheel.



Deserve "A" grade!

– Oleg Bulichev

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📍 @Lupasic

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